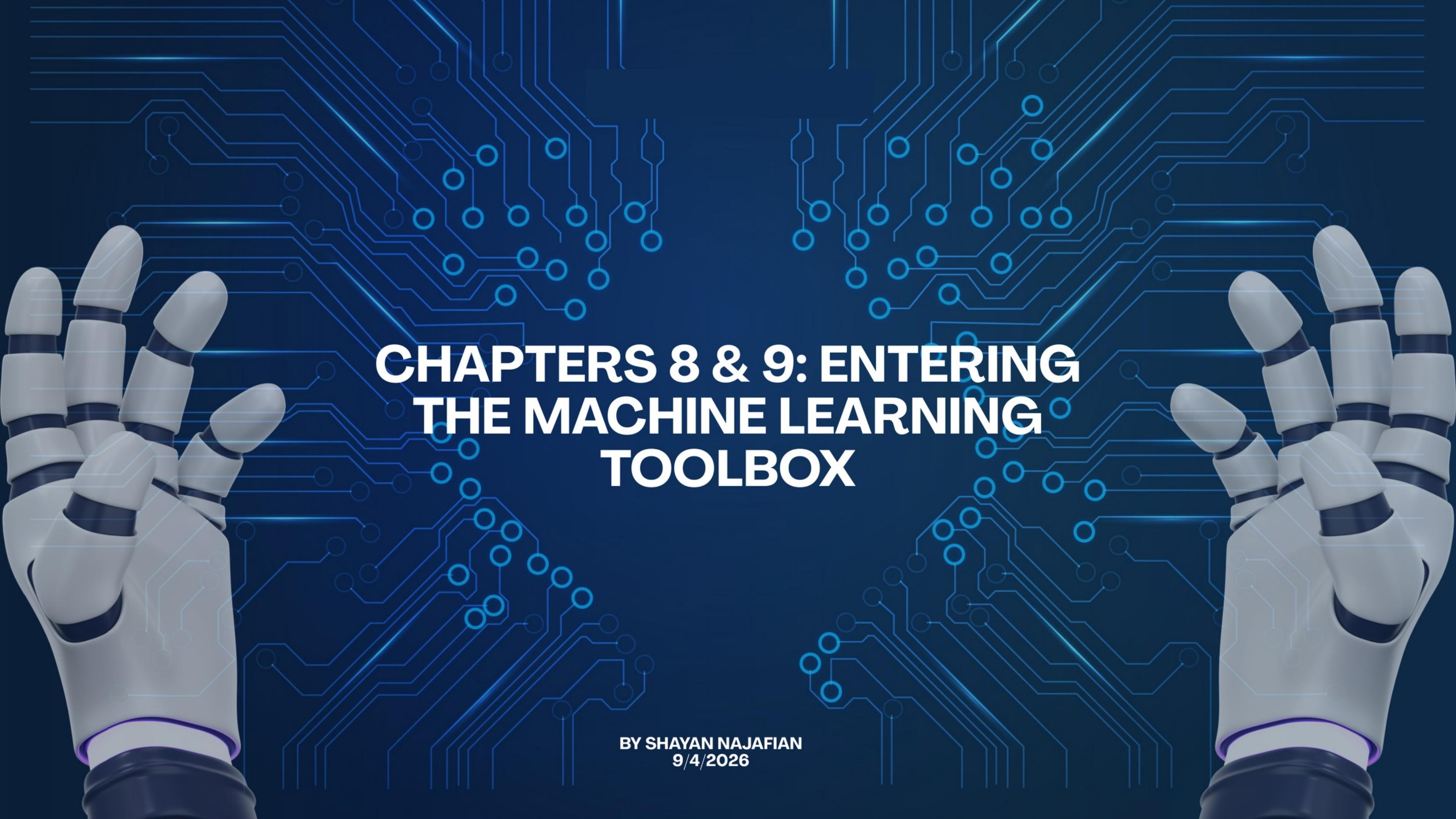




CHAPTERS 8 & 9: ENTERING THE MACHINE LEARNING TOOLBOX

BY SHAYAN NAJAFIAN
9/4/2026

THE MACHINE LEARNING MAP

- Unsupervised Learning
Find hidden structure
- Supervised Learning
Learn from known answers

CHAPTER 8: SEARCH FOR HIDDEN GROUPS

Dimensionality Reduction
Clustering

CUSTOMER SEGMENTATION WITHOUT LABELS

Imagine 10,000 customers.

For each customer, we know:

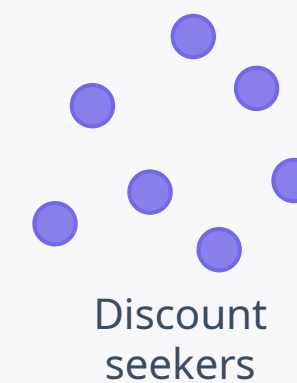
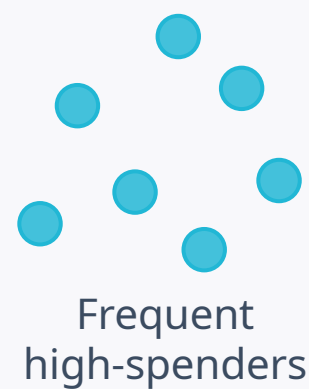
Age • Income • Purchase frequency

Average spending • Product preferences

But we have no predefined customer types.

CAN THE DATA REVEAL CUSTOMER TYPES?

No labels like “premium” or “discount seeker.”
Only behavior and demographics.
The algorithm finds groups; we name them afterward.



Unsupervised learning starts with data, not answers.

ROWS, COLUMNS & DIMENSIONS

Rows = observations
Columns = features
Number of features = dimensionality

Customer	Age	Income	Purchases	Spending
A	24	\$35k	12	\$240
B	42	\$90k	3	\$850
C	31	\$55k	8	\$410

WHY HIGH DIMENSIONALITY IS HARD

More features do not always mean more insight.
Many variables may be redundant, noisy, or hard to visualize.
High-dimensional data is difficult to explore, compare, and model.

We need a smaller view that preserves the important structure.

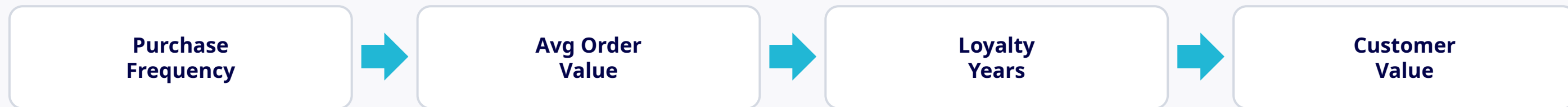
THE CURSE OF DIMENSIONALITY

As dimensions increase:
Data becomes more sparse.
Distance becomes less meaningful.
More data is needed.
Models can overfit more easily.

The space grows faster than our data.

DIMENSIONALITY REDUCTION

Many original features → fewer new features
Goal: keep most of the information while simplifying the data.

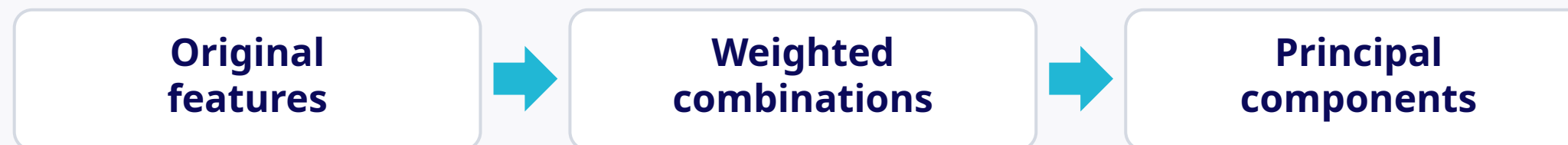


PCA: TECHNICAL INTUITION

PCA searches for new directions that capture maximum variance.

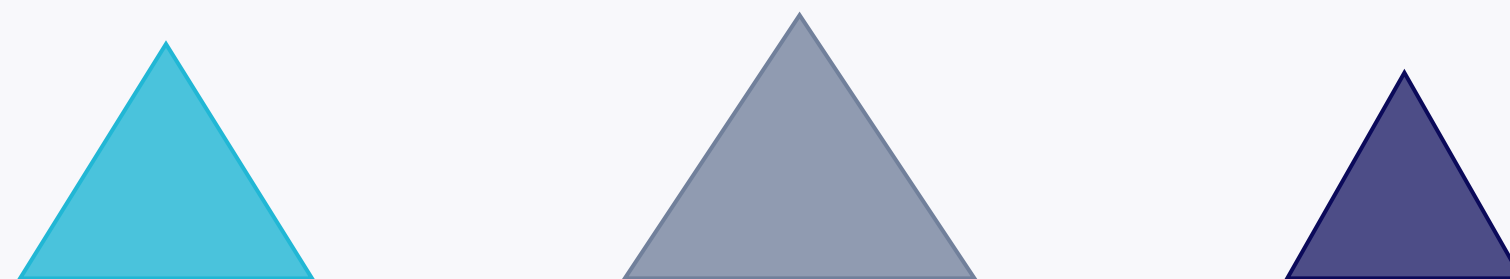
PC1 captures the most variance.

PC2 captures the next most, while being uncorrelated with PC1.



PCA AS A CAMERA ROTATION

Same object. Different view.
Some angles preserve more information than others.
PCA searches for a useful view of the data.



PCA EXAMPLE: ATHLETIC ABILITY

30 athletic measurements may contain fewer hidden dimensions:
Strength, Speed, Stamina, Health.

PCA discovers these patterns through correlations among features.

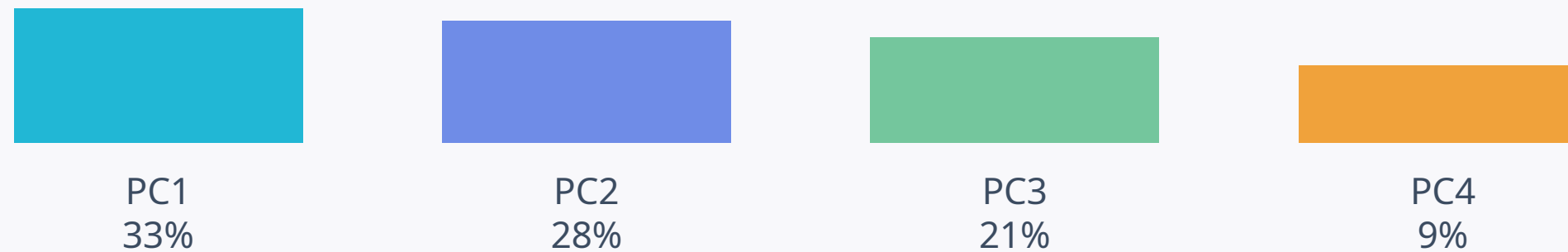
PCA WEIGHTS

A principal component is a weighted combination of features.
Weights show how much each original feature contributes.

$$\text{Strength} \approx 0.6(\text{Push-ups}) + 0.5(\text{Deadlifts}) + 0.4(\text{Sit-ups}) + \dots$$

EXPLAINED VARIANCE

The first few components should carry most of the useful spread.
In the book's athlete example: 30 features → 4 components → ~91%.



PCA CAN MISLEAD TOO

High variance $\neq$ important.
Components may not have clean meanings.
Scaling matters.
Humans decide how many components to keep.

CLUSTERING: GROUP THE ROWS

PCA condenses columns.

Clustering groups rows.

Customers → segments. Stores → regions. Athletes → types.

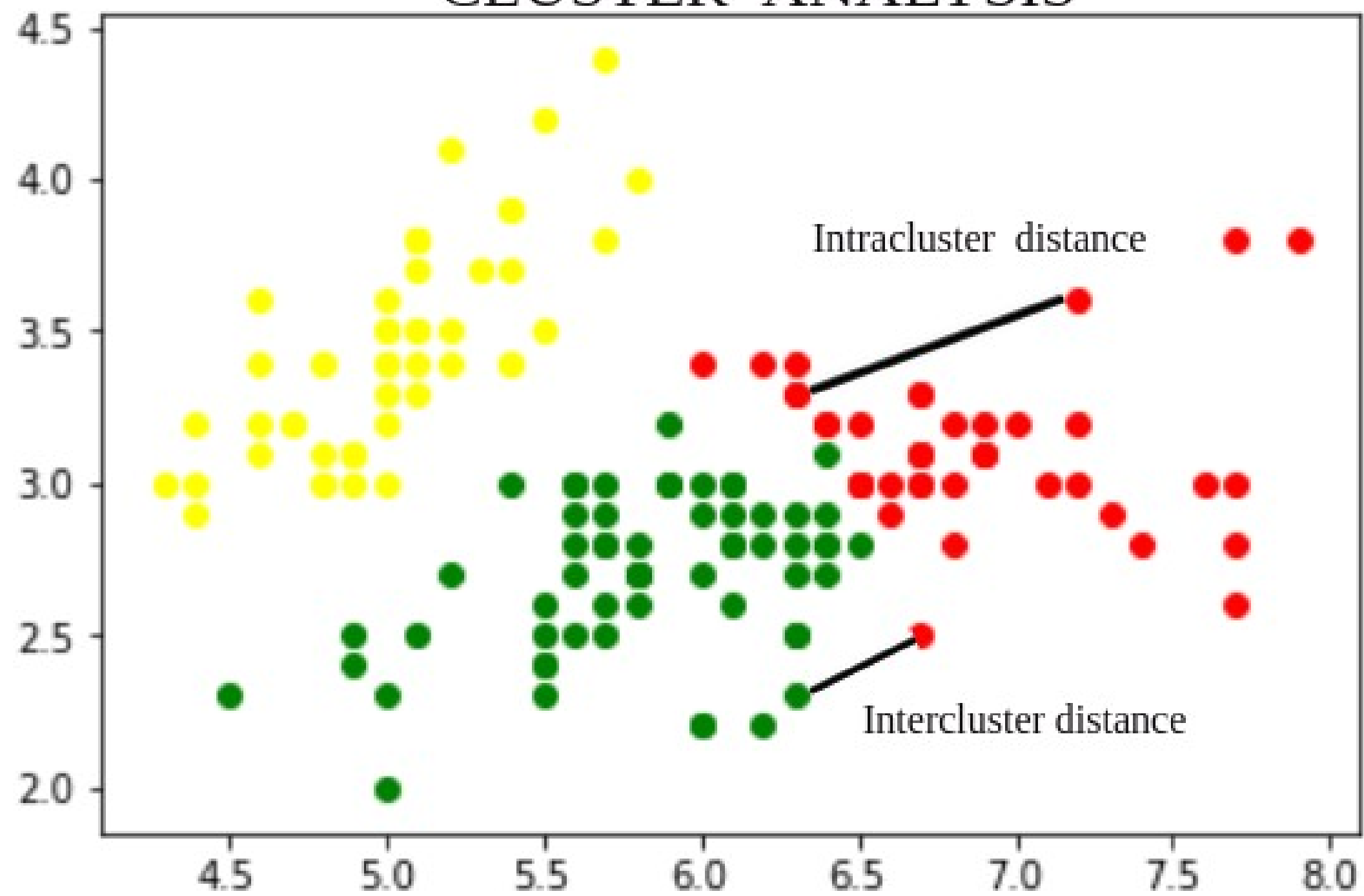
WHAT DOES “SIMILAR” MEAN?

Before clustering, define what “close” means.

Age? Income? Behavior? Location?

Different definitions of similarity can create different groups.

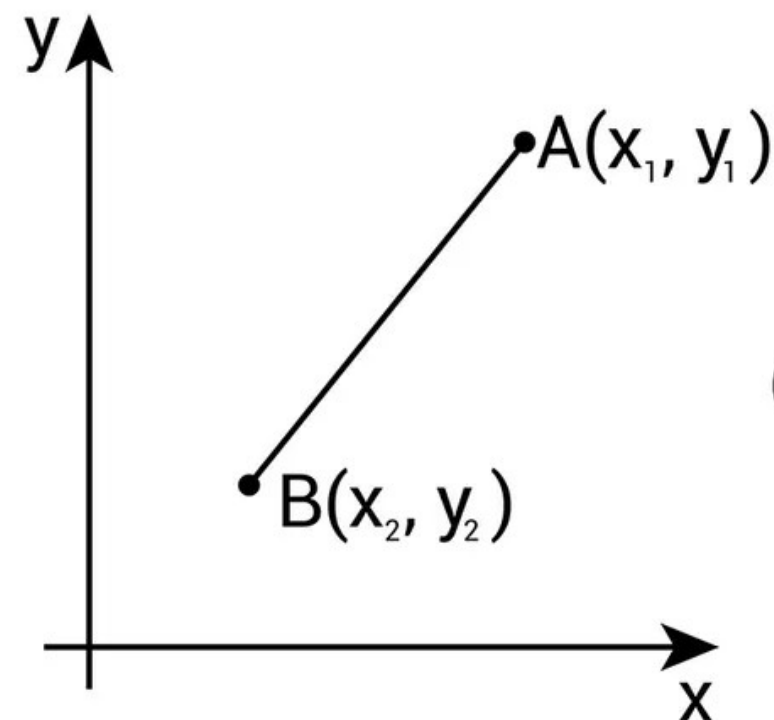
CLUSTER ANALYSIS



DISTANCE: JUST ENOUGH MATH

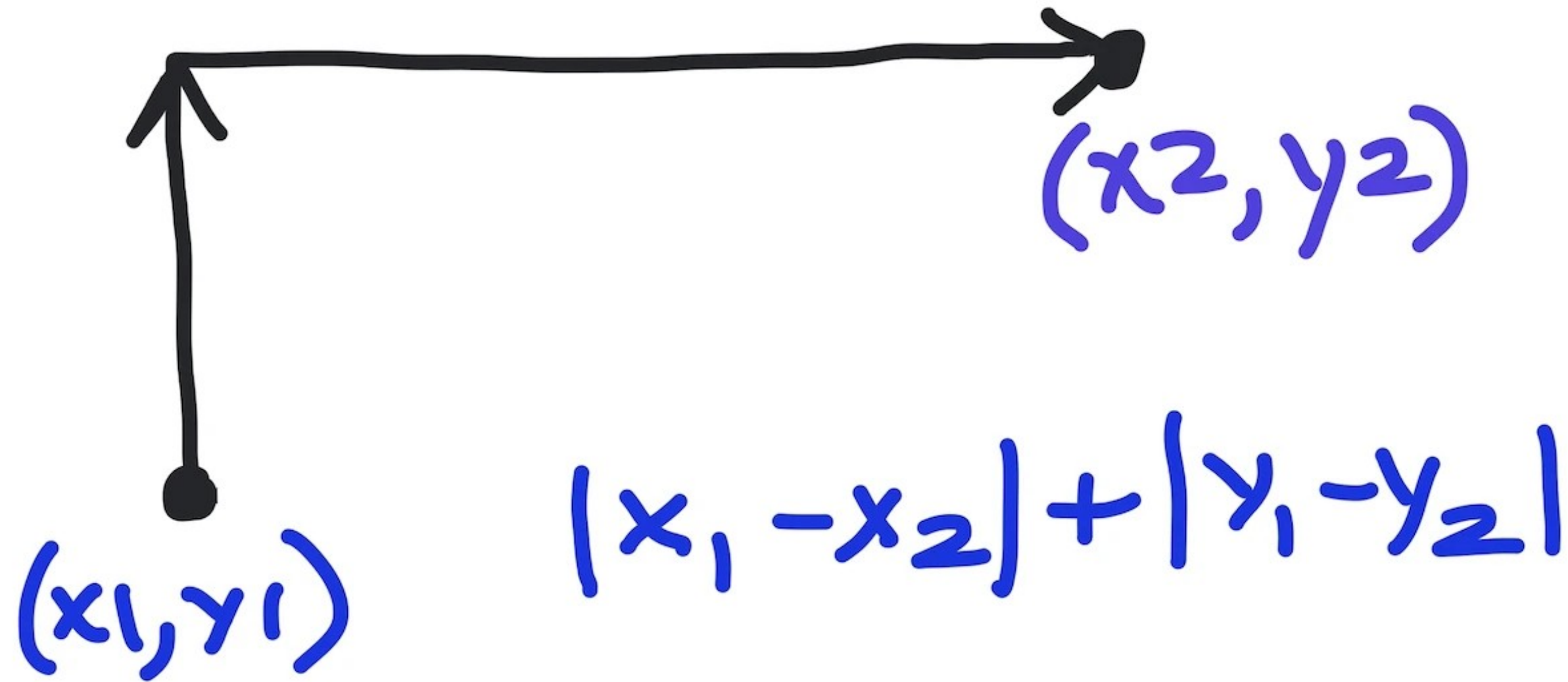
Clustering needs a numerical definition of closeness.
One common choice is straight-line distance.

Distance Formula

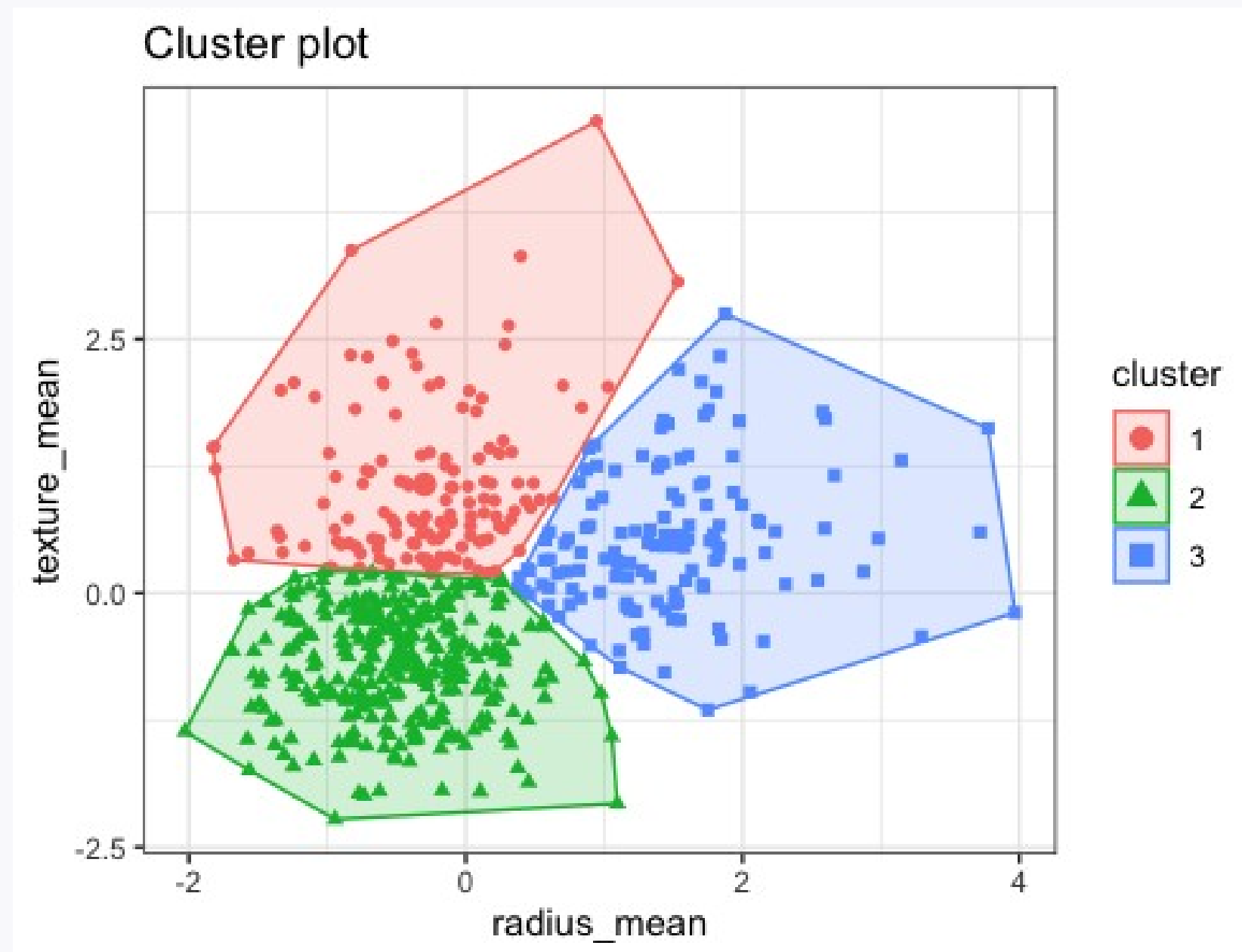


$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

manhattan distance



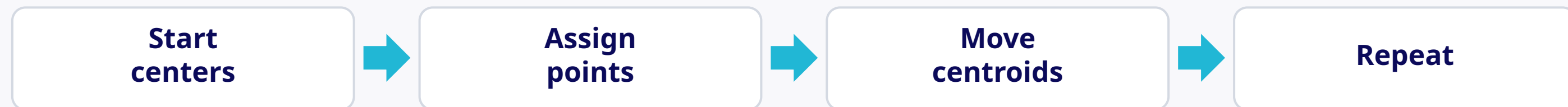
K-MEANS CLUSTERING



k = number of clusters.
Example: 200 stores, k = 6.
Goal: six useful geographic groups.

HOW K-MEANS WORKS

Pick k starting centers.
Assign points to the nearest center.
Move each center to the average of its group.
Repeat until the clusters stabilize.



SCALING CAN CHANGE THE CLUSTERS

Age = 36

Children = 3

Income = 100,000

Which feature dominates the distance calculation? Income.

Standardization puts features on comparable scales.

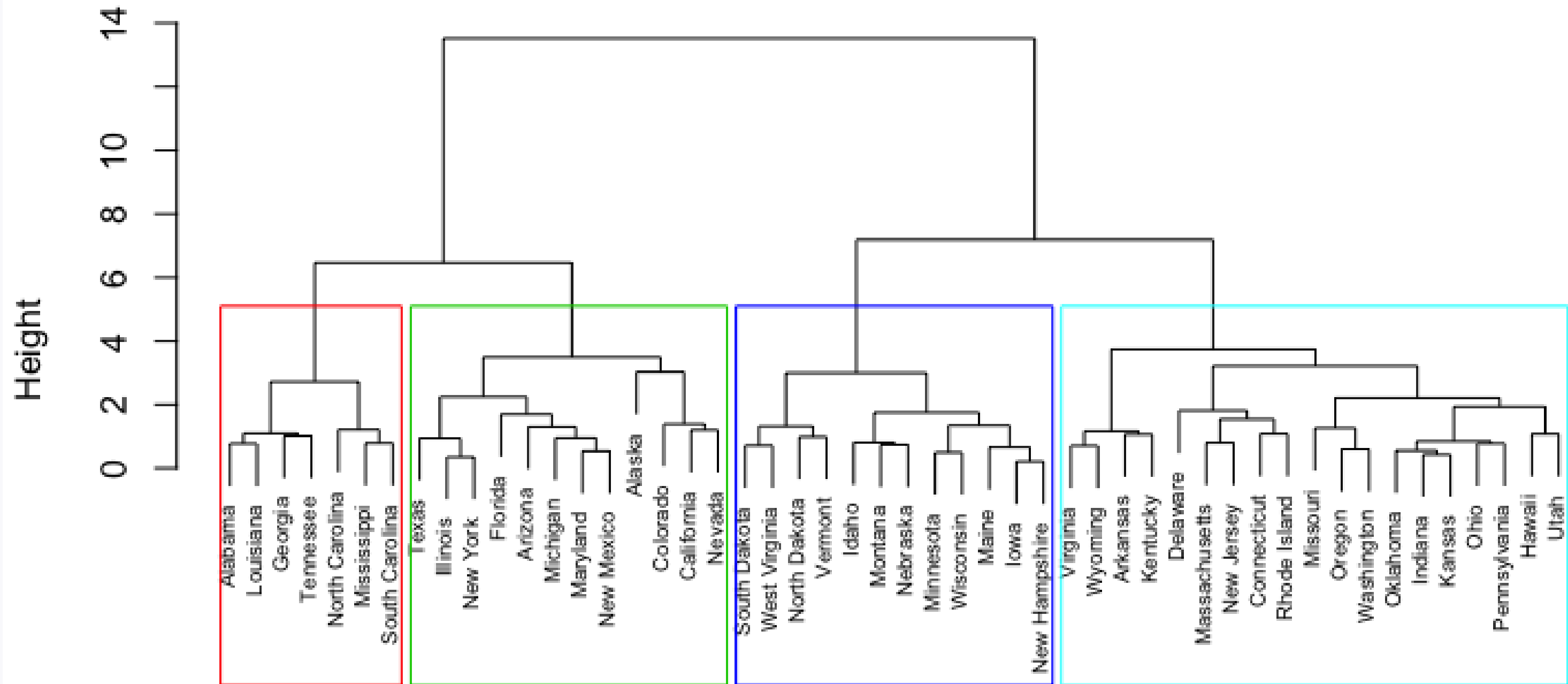
HIERARCHICAL CLUSTERING

Start small.

Combine similar observations into groups.
Then combine smaller groups into broader groups.

Example: contemporary jazz + classic jazz → jazz.

Cluster Dendrogram



CHAPTER 8 TAKEAWAY

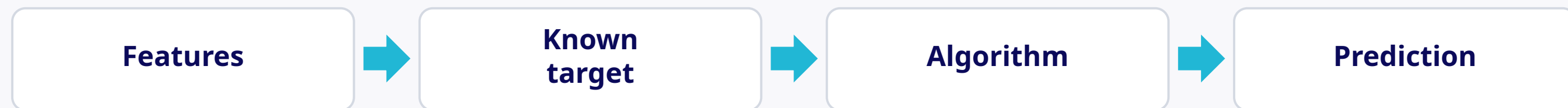
“Unsupervised” learning still requires supervision.
Humans choose features, scaling, distance, k, components, and interpretation.
The algorithm finds patterns; we judge whether they are useful.

CHAPTER 9: REGRESSION MODELS

Now we move from hidden groups to known answers.
Regression learns from historical examples to predict a number.

SUPERVISED LEARNING

Features + known target → learn relationship → predict new target.
The model learns from examples where the answer is already known.



REGRESSION VS. CLASSIFICATION

Regression

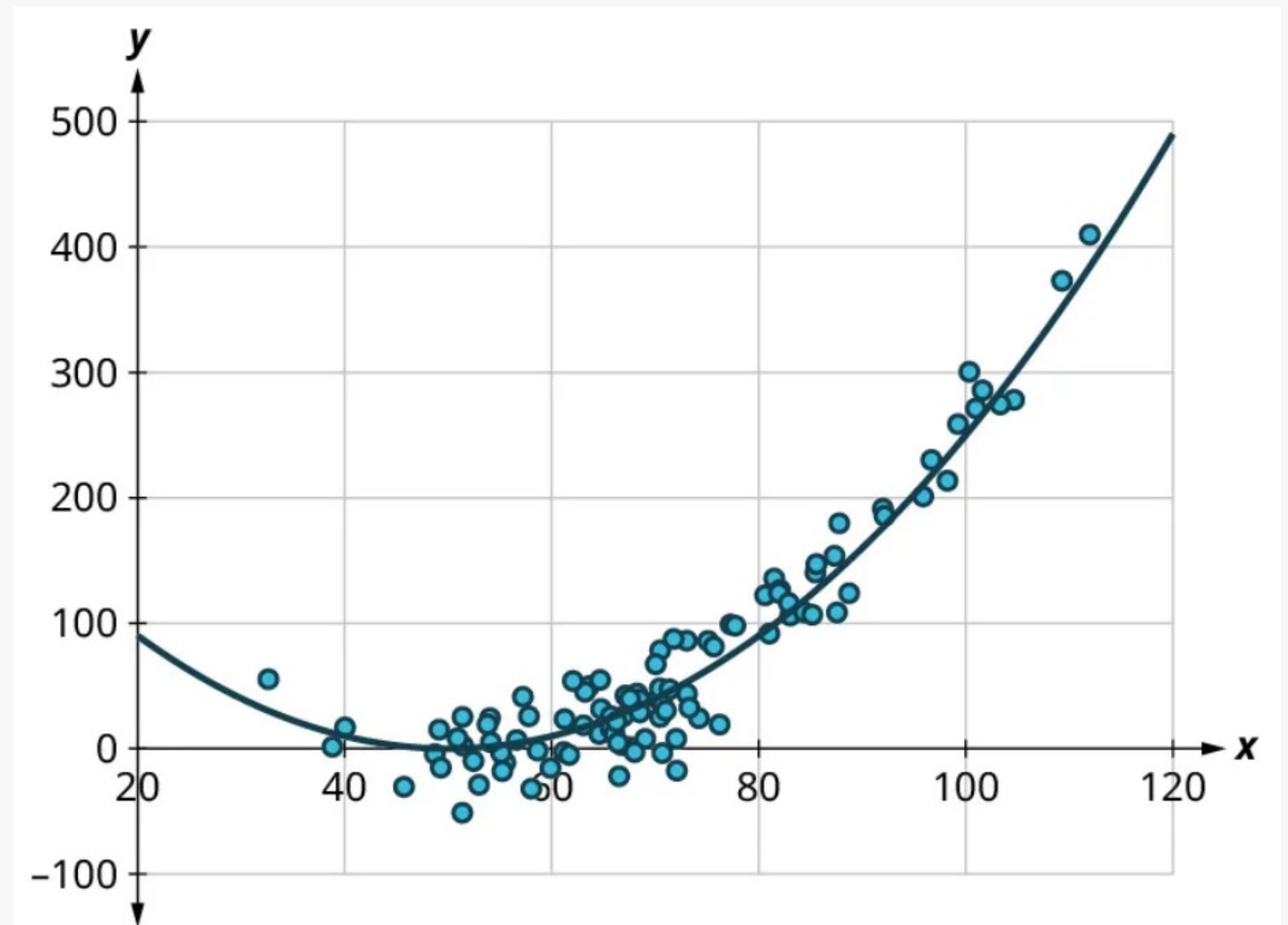
Numeric target
House → \$350,000
Sales forecast

Classification

Category target
Email → Spam
Fraud / Not fraud

LINEAR REGRESSION

$\hat{y} = b_0 + b_1x$
 x = feature, $\hat{y}$ = prediction
 b_0 = intercept, b_1 = slope

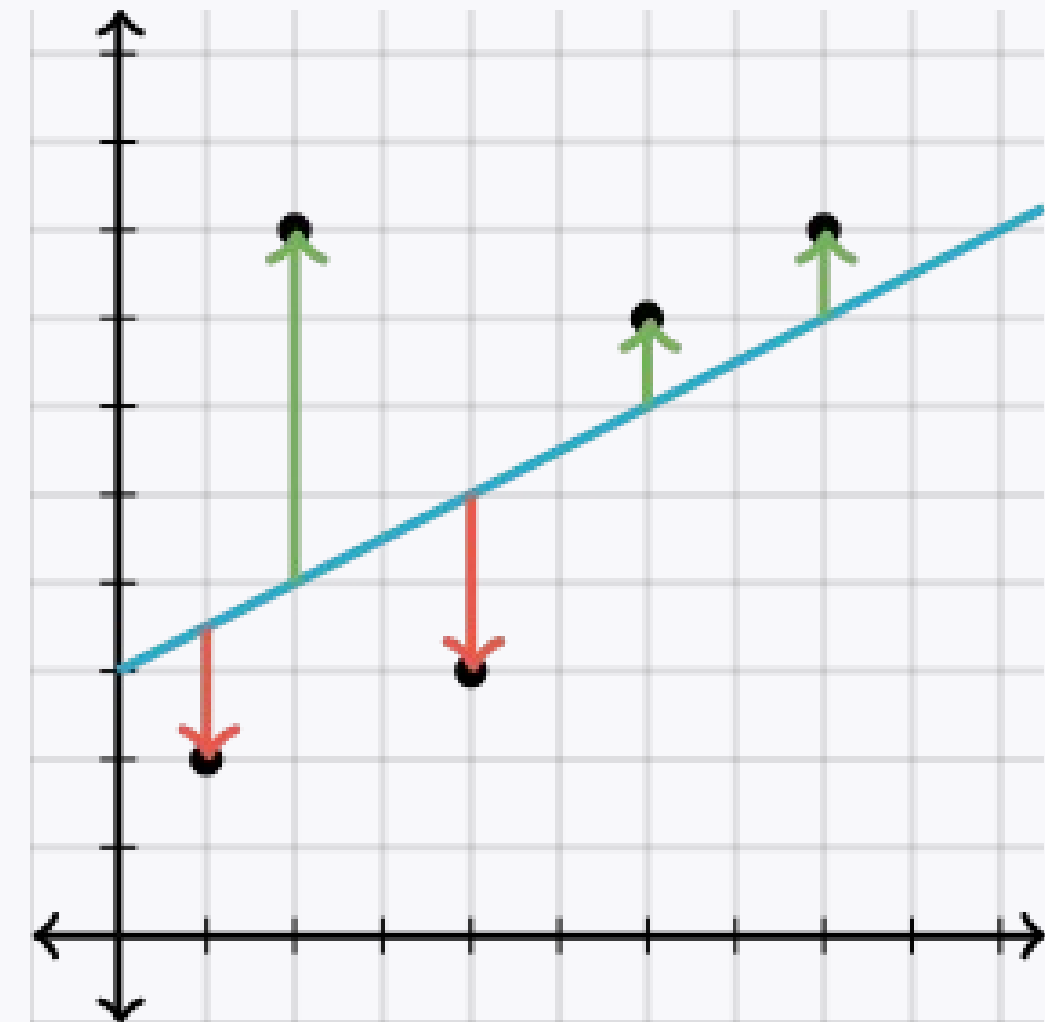


WHICH LINE IS BEST?

Many lines can look reasonable.
Linear regression chooses the line with the smallest prediction errors.
This is better than drawing a line by eye.

RESIDUALS & LEAST SQUARES

Residual = Actual - Predicted
Linear regression minimizes total squared residuals.
 $SSE = \sum((\text{Actual} - \text{Predicted})^2)$



WHY SQUARE THE ERRORS?

Without squaring: $+4 + (-4) = 0$
But both predictions were wrong.
With squaring: $4^2 + (-4)^2 = 32$

WHAT DOES A COEFFICIENT MEAN?

$$\text{Sales} = 1.03 \times \text{Temperature} - 71.07$$

$$\text{Coefficient} = 1.03$$

One-unit increase in temperature is associated with about \$1.03 higher predicted sales.

Associated with is safer than immediately saying caused by.

R-SQUARED

R^2 tells us how much variation the model explains relative to a baseline.
Example: $R^2 \approx 0.788$ means about 78.8% explained in the observed data.
High R^2 is not automatically a good causal model.

MULTIPLE REGRESSION

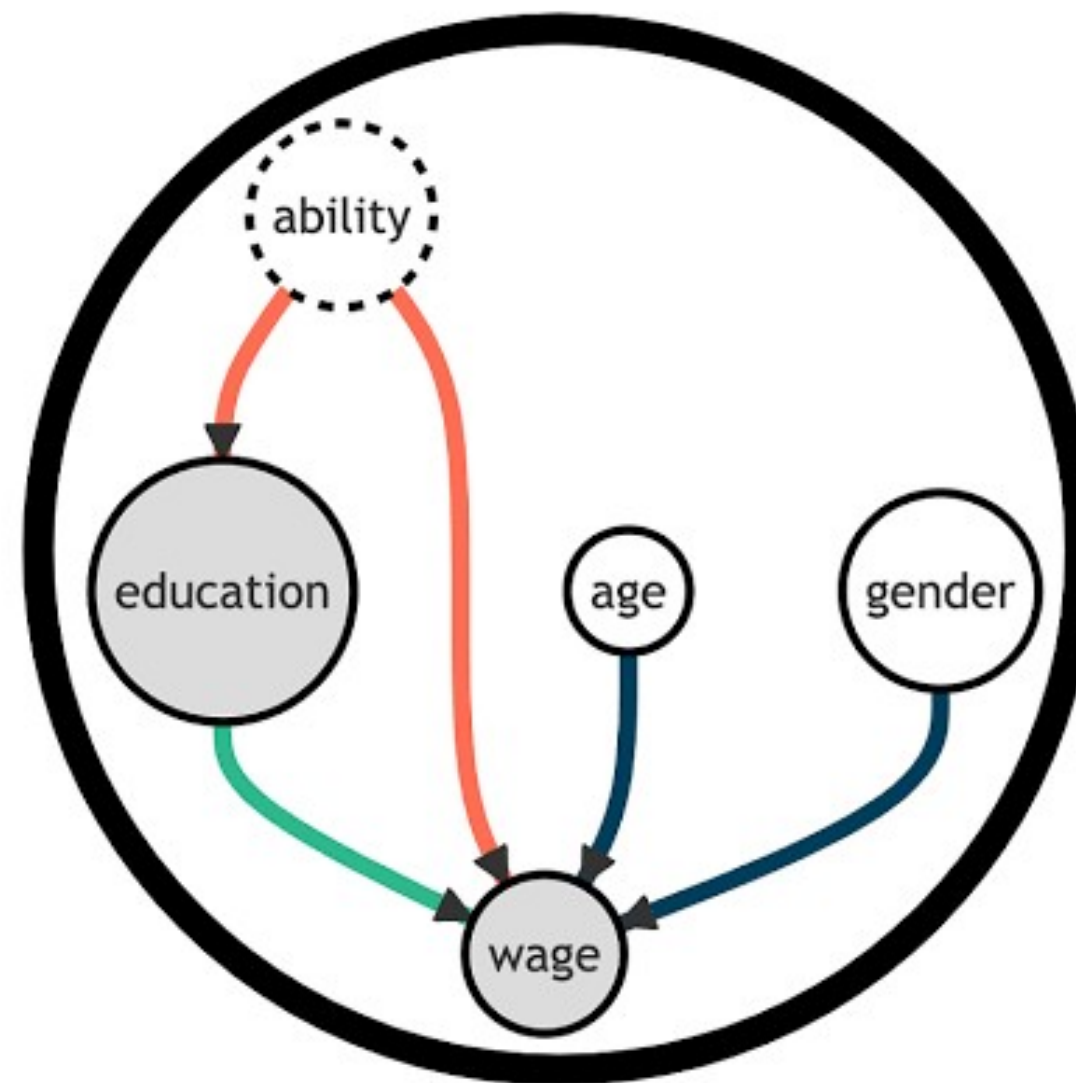
House Price = $b_0 + b_1(\text{Lot Area}) + b_2(\text{Year Built}) + b_3(\text{Floor Area}) + \dots$

The key phrase: holding other included variables constant.

FAILURE #1: OMITTED VARIABLES

Model finds: Shoe Size → Reading Ability.
But age affects both shoe size and reading ability.
A model cannot learn from a variable we did not include.

FAILURE #1: OMITTED VARIABLES



FAILURE #2: MULTICOLLINEARITY

Temperature in °C and temperature in °F contain almost the same information.

Regression struggles to separate their individual effects.
This makes coefficients hard to interpret.

FAILURE #3: DATA LEAKAGE

Goal: predict sale price before the house sells.

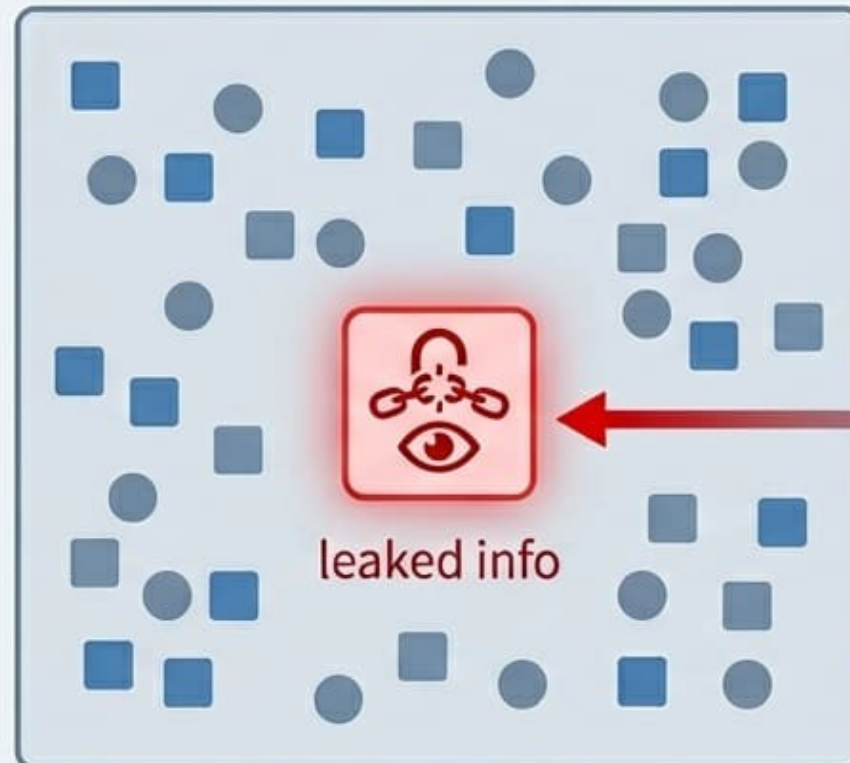
Feature: initial offer.

Problem: was that information available at prediction time?

Leakage makes the model look smart because it peeked at the future.

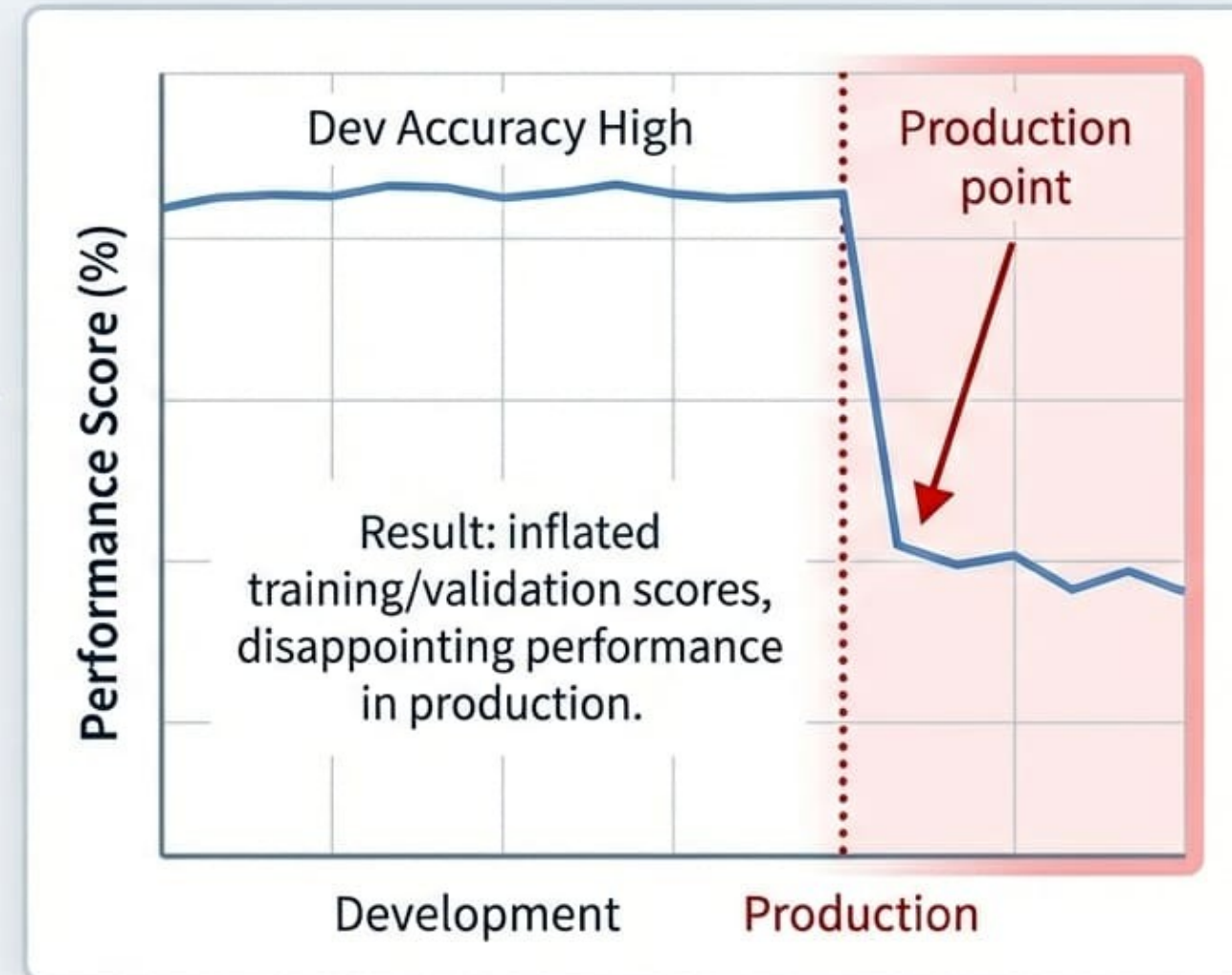
FAILURE #3: DATA LEAKAGE

Training Dataset Box



Data leakage: your model sees information during training that it would never see when making real predictions.

Performance Chart



Timeline (Development → Production)

FAILURE #4: EXTRAPOLATION

Training temperatures: 80°F - 86°F.

Question: what happens at 0°F?

Model answer: negative lemonade sales.

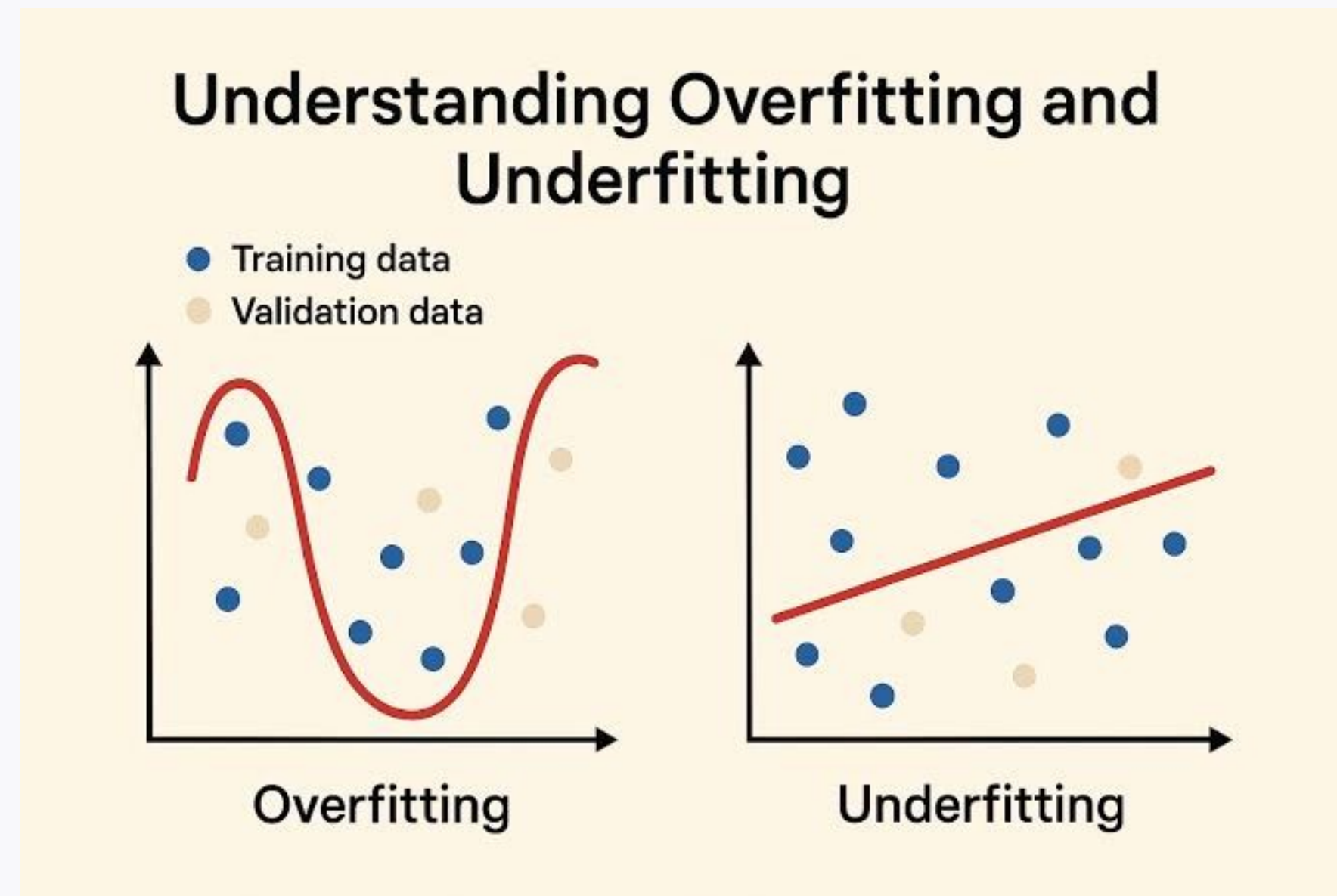
Models do mathematics. They do not have common sense.

FAILURE #5: NOT EVERYTHING IS LINEAR

Linear regression assumes a straight-line relationship is useful.
Some relationships are curved, exponential, seasonal, or threshold-based.
Don't fit a straight line through a banana.

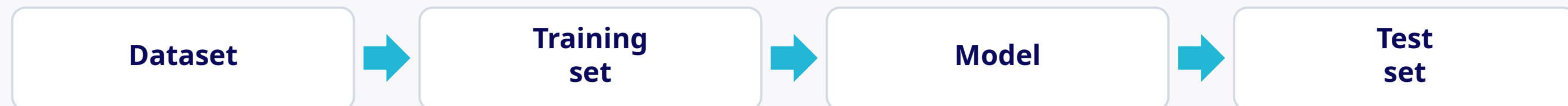
FAILURE #6: OVERFITTING

Perfect training fit $\neq$ good prediction.
An overfit model memorizes noise instead of learning a general pattern.



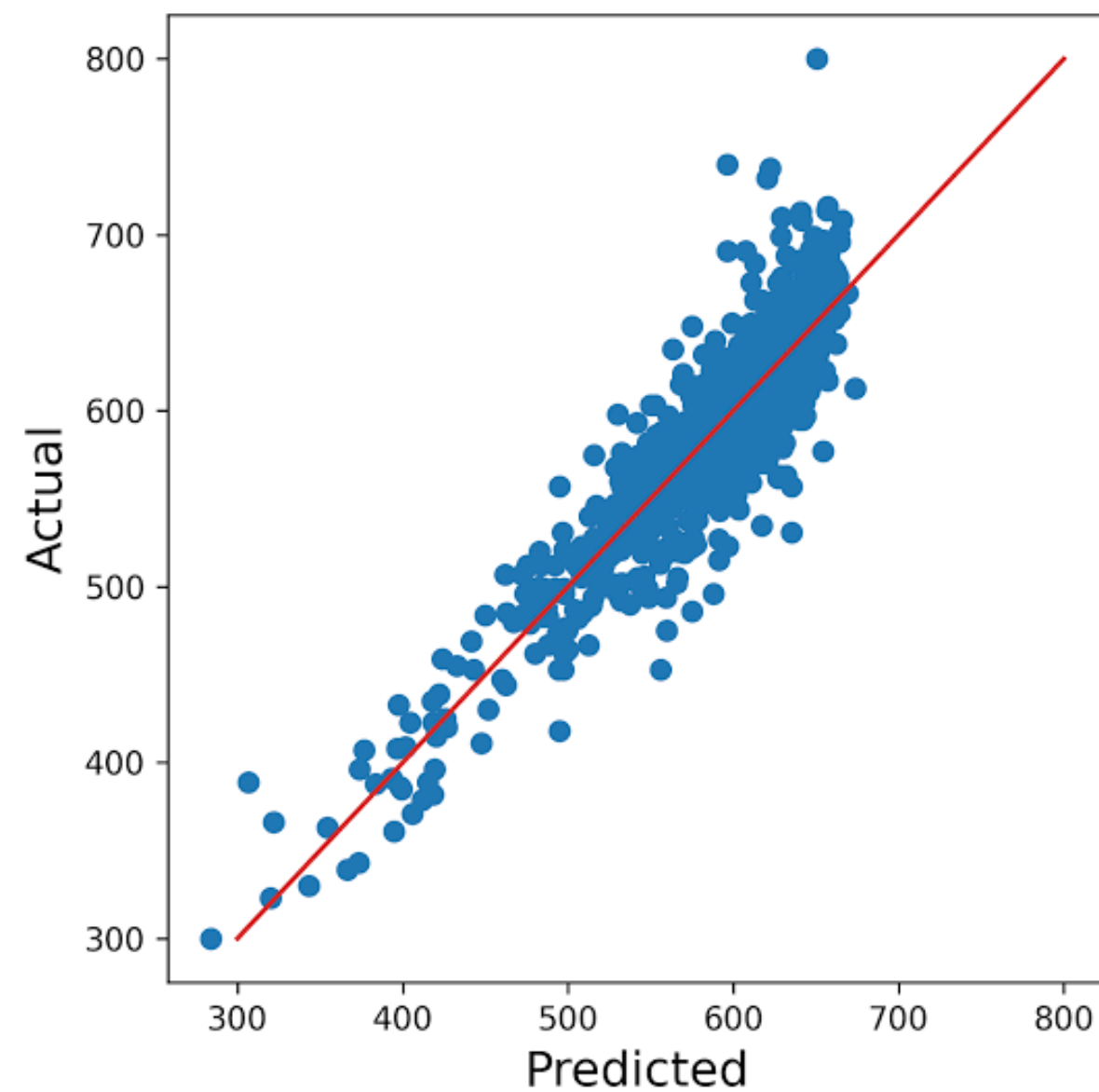
TRAINING SET VS. TEST SET

Training set: build the model.
Test set: evaluate on data the model has never seen.
The real question is generalization.



ACTUAL VS. PREDICTED

For each observation, compare actual value with predicted value.
Good predictions should be close to the diagonal.
Do not trust R^2 alone. Look at the errors.



EXPLAINING OR PREDICTING?

Explanation asks: why does X relate to Y?
Prediction asks: how accurately can we predict Y?
These are related goals, but not the same goal.

THE CONNECTION BETWEEN CHAPTERS 8 & 9

Chapter 8: no known answers → find structure.
Chapter 9: known answers → learn relationships.
Together, they give the basic map of machine learning.

CONCLUSION: ALGORITHMS CALCULATE, HUMANS JUDGE

Prediction $\neq$ causation.

Good fit $\neq$ good model.

Machine learning moves human judgment to different parts of the process.

THYNK UNLIMITED

**THANK
YOU.**

